



NoShot Video Browser at VBS2021

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Abstract. We present our NoShot Video Browser, which has been successfully used at the last Video Browser Showdown competition VBS2020 at the MMM2020. NoShot is given its name due to the fact, that it neither makes use of any kind of shot detection nor utilize the VBS master shots. Instead videos are split into frames with a time distance of one second. The biggest strength of the system lies in its feature “time cache”, which shows results with the best confidence in a range of seconds.

Keywords: Video retrieval · Interactive video search · Video analysis

1 Introduction

As a participating system for the 9th edition (VBS2020) of the annual interactive video retrieval challenge known as the Video Browser Showdown [4, 5, 8, 10], the NoShot Video Browser provides a simple and easy-to-use web UI for searching in big video collections, such as the competition’s currently employed V3C1 dataset [1, 9] including approx. 1 000 h of video. The system has a very low query and user interaction response time and can be used on every modern browser.

It specifically was developed at the AAU Klagenfurt in order to provide a simpler UI approach as the already existing, feature-rich diveXplore [2, 3, 6, 11, 12] system. Yet possessing by far fewer features, NoShot, however, shines through its clean easy-to-use UI, which especially has many advantages for the challenge’s novice session: for determining a final team score the VBS traditionally features two sessions – one for expert users, where usually the systems are used by their developers and one for novice users, where novice users not familiar with the systems compete against each other. In the latter case, therefore, it is paramount to provide self-explaining, simple interfaces in order to maximize the success rate of a system.

In addition to basic UI improvements and code refactorings, new features such as a novel brightness filter is added for improving search results. Through extensive testing and optimizations, We expect the system to perform well in VBS2021.

2 NoShot Video Browser

The NoShot system has already been used for the Video Browser Showdown 2020. It achieved the 7th place. As the system has been used in a competition for the first time, we were able to test the features of our system.

Every video in the dataset is split into keyframes with one second of time distance. NoShot uses YOLO 9k [7] to analyze every of those frames and to label them with up to five of the 9000 classes. The classes are organized into a hierarchy, so we additionally store the parent class id. This step takes a lot of time and processing power and is therefore done beforehand. The classification data is loaded into an Apache Solr¹ database. Apache Solr processes a search query with the VBS2020 data in under 100 milliseconds. It provides automatic caching and pagination, which further speeds up search queries. A search API written in Node.js² gets search requests of the frontend and builds a Solr search query. This query is sent to the Solr server via a http json request. The front end is written in JavaScript and uses Vue.js³ as the UI framework. The whole NoShot system is able to process a search request in a fraction of a second.

3 Time Cache

A core feature of the NoShot Video Browser is its time cache. This feature allows the system to show only the best keyframes of a certain time period measured in seconds. The best keyframes are found by sorting by the confidence returned from the YOLO 9000 CNN. In a preprocessing step the best keyframes for all categories are cached in Apache Solr. Furthermore, we create multiple caches (for 30, 60 and 180 s). By caching the results in Apache Solr, we ensure, that every search query has a low response time. At the VBS2020, especially the longest cache (180 s time cache) was useful, as we could successfully remove duplicate results. If a certain scene is shown for a long time only one frame per category is displayed. As this frame has a high confidence for that category, the detected object is often shown fully on the frame.

4 The NoShot GUI

The front end of the NoShot Video Browser is shown in Fig. 1. To enable the user to switch between several search views, the front end provides a tab view. The search field under the tab view is designed big to help the user to start the search query fast. After hitting the return key, a new search window appears. In this window the search keyword can be changed. While the user is typing, the search window refreshes. To help the user to find the best fitting of all 9000 classes, we show a autocomplete tool. The search window uses pagination for

¹ <https://lucene.apache.org/solr/>.

² <https://nodejs.org>.

³ <https://vuejs.org>.

big result lists. The cache input sets the length of the time cache (1, 30, 60 or 180 s). The range input shows the frames before and after each result keyframe. The Video view is accessed via an overlay window on top of the tab view and provides video playback functionality. It is opened by holding the control key while clicking on a frame. The video playback position is set to the frame clicked. By holding the shift key while clicking on a frame, the video is opened in the video summary window. In this window the frames of a video are shown. With the cache input the video summary window can be set to only show a fraction of the video frames.

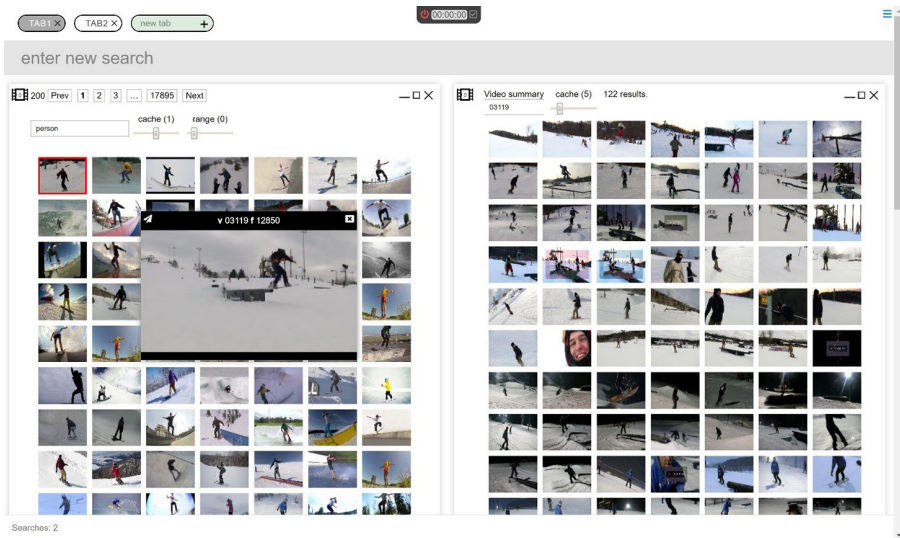


Fig. 1. The NoShot Video Browser

5 Improvements

In comparison with the last version of NoShot, we add a new feature called brightness filter. We add a preprocessing step to analyze the brightness of all keyframes in our database. The brightness filter separates the frame in several sections of the same size. The result of this analysis is stored in the Apache Solr database. The user of the system gets a new input field to specify a section, where retrieved result keyframes should be bright. The system filters all keyframes according to the selected section. Brightness is defined relative to the other sections of the same frame, thus for some frames the brightest regions could appear light gray, while for other frames they appear white. To improve the usability, we include the adjacent areas of a keyframe in the brightness search.

6 Conclusion

We introduce our NoShot Video Browser for participating in VBS2021, which is a web-technologies based interactive video search tool. We are confident, that our time cache feature is beneficial for users of the system. We further improved the system, by adding a brightness search filter. By combining the keyword search with the time cache and the brightness filter, the user can reduce the amount of results significantly. The most important factor for the new feature is its fast response time.

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Exquisitor at the Video Browser Showdown 2021: Relationships Between Semantic Classifiers

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Abstract. Exquisitor is a scalable media exploration system based on interactive learning, which first took part in VBS in 2020. This paper presents an extension to Exquisitor, which supports operations on semantic classifiers to solve VBS tasks with temporal constraints. We outline the approach and present preliminary results, which indicate the potential of the approach.

Keywords: Interactive learning · Video browsing · Temporal relations

1 Introduction

The Video Browser Showdown (VBS), now in its 10th anniversary edition, has emerged as an important vehicle for the evolution of the multimedia field [5]. During VBS, researchers are given a series of never-before-seen task descriptions, based on a collection of 7,475 video clips [9], and asked to interactively retrieve either one specific video segment or multiple relevant segments, depending on the task type. VBS allows researchers working on media exploration and search tools to apply their techniques in a realistic setting and better understand the pros and cons of both the underlying techniques and the interfaces. The lessons learned during the competition can then inspire new methods and further research. In addition, the competitive setting makes for an exciting event where the ranking of systems can also give hints to their usability and applicability.

Exquisitor, a prototype media exploration system based on interactive learning, took part for the first time in VBS 2020, where it placed 5th out of 11 systems [2]. The goal of Exquisitor, as applied to VBS, is to build a semantic classifier for the information need represented in each task, and use that classifier—along with metadata filters and a video timeline explorer—to solve the task. Exquisitor uses the video segmentation supplied with the VBS collection and represents each video segment independently by semantic features

derived from its keyframe. When building the semantic classifier, Exquisitor suggests keyframes to the user and asks for feedback on those suggestions. Once the user spots a potentially relevant keyframe, the video explorer can then be used to explore the actual content and internal structure of the full video clip.

For many VBS tasks, the task description applies to more than one video segment, often focusing on different semantic concepts in different segments, and sometimes providing an explicit temporal relationship. Unsurprisingly, therefore, all the strongest VBS competitors provide temporal queries as a major technique [4, 6, 7, 10]. Since video segmentation tends to split the video by semantic concepts, a classifier built to find one segment may not find the other, and the system should provide support to utilise the relationship between concepts in video segments.

In this paper, we present a new version of Exquisitor, where the major extension is the support for utilising relationships between semantic classifiers. While each semantic classifier is developed in the same manner as before, using independent video segments, the results of two semantic classifiers can now be combined in various ways, with an optional temporal relationship specification. In this paper we briefly outline the method and interface for combining two semantic models and show how two models combined could be used to solve two VBS 2020 tasks, one of which the team failed to solve during the competition. We will present and evaluate the methods in more detail in a later publication.

2 Exquisitor

Exquisitor is a user relevance feedback approach capable of handling large scale collections in real time [3, 8]. The Exquisitor system used for VBS consists of three parts: (1) a web-based user interface for receiving and judging video suggestions; (2) an interactive learning server, which receives user judgments and produces a new round of suggestions; and (3) a web server which serves videos and video thumbnails. Due to the computational efficiency of the system, all three components can run locally on a laptop.

Exquisitor Server: Exquisitor is fueled by a semantic model that combines interactive multimodal learning with cluster-based indexing. Each keyframe in each modality is represented by an efficient representation containing the most important semantic features, compressed using an index-based method [11]. This representation is further clustered using a cluster-based indexing approach [1]. When building a semantic classifier C , a linear SVM classifier is iteratively refined based on user interactions (positive and negative examples). In each round of interaction, the resulting separating hyperplane forms k -farthest neighbour queries posed to the cluster-based indexes. Finally, late fusion is performed on the retrieved results, to produce the 25 top-ranked results to suggest to the user.